

Software Testing

QP Code : 04550

(2½ hours)

Total Marks: 75

- N. B.: (1) All questions are compulsory.
(2) Make suitable assumptions wherever necessary and state the assumptions made.
(3) Answers to the same question must be written together.
(4) Numbers to the right indicate marks.
(5) Draw neat labeled diagrams wherever necessary.
(6) Use of Non-programmable calculators is allowed.

1. Answer any two of the following: 10
a. State the phases of Fundamental Test Process. Explain Planning and Control.
b. What is Quality? Explain Quality Viewpoints for producing and buying software.
c. Write a short note on Psychology of Testing.
d. State principles of Software Testing. Explain any four in detail.

2. Answer any two of the following: 10
a. Explain Lifecycle Testing for Iterative model and RAD model.
b. What is Integration Testing? List and explain the approaches Integration Testing.
c. Explain Triggers for Maintenance Testing.
d. Explain Conformance directed Testing Levels.

3. Answer any two of the following: 10
a. List the phases of Formal Reviews. Explain any four phases in detail.
b. Explain the characteristics and objectives of :-
 i) Walkthrough
 ii) Inspection
 iii) Audit
 iv) Technical review
 v) Peer Review
c. Explain the factors influencing successful performance of Reviews.
d. List and explain any two types of Static Analysis using Tools.

4. Answer any two of the following: 10
a. Explain how Equivalence Class Partitioning differs from BVA.
b. Explain Experience based testing and its types.
c. Describe State Transition based testing with an example.
d. Define Cyclomatic Complexity:
 Draw a Program Graph to Graphically represent the given Code and find the Cyclomatic complexity.

```
Dim i as Integer
Dim n as Integer
i=1
n=1
while(i<=5)
{
    n=n*i
    i=i+1
}
output n
```

5. Answer any two of the following:

- a. What is Risk Based Testing? How is product risk different from project risk?
- b. Explain incident report lifecycle with the help of a diagram.
- c. Define Test Control. Explain configuration management in detail.
- d. Explain Test Organization. What are the factors affecting test efforts?

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6. Answer any two of the following:

- a. Explain the potential benefits and risks of test automation and tools support for testing.
- b. State the objectives of Proof-of-Concept or Piloting phase for tool evaluation.
- c. Describe the features of Test Design tools and Data Preparation tools.
- d. Explain in detail how tools provide support for Test Specification activity.

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7. Answer any three of the following:

- a. Define the terms Risk, Error, Defect, Failure and Quality.
- b. Explain V- Model of testing with the help of a diagram. Explain its advantages.
- c. Define Static Testing. Explain the significance of Static Testing.
- d. Justify "100% Decision Coverage implies 100% Statement Coverage" giving an example.
- e. Explain various types of Test Strategies.
- f. Discuss the features of Test Execution Tool. List the considerations for implementing these tools.

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